General relativity HW-10

Problem 1. Consider FLRW (or Friedmann) metric with a non-zero spatial curvature. Prove that it is conformally flat.

Solution:
This solution partly refers to https://arxiv.org/pdf/hep-th/0702139
We know that the Friedmann metric is

$$ds^2 = -dt^2 + a^2(t)(\frac{dr^2}{1-kr^2} + r^2 d\Omega^2)$$

For $k = -1$,we firstly let $dt = a(\tau)d\tau$.Then the metric becomes

$$ds^2 = a^2(\tau)(-d\tau^2 + \frac{dr^2}{1+r^2} + r^2 d\Omega^2)$$

Secondly,let's consider how to make the spatial part 'flat'.We may at the first glance consider the substitution $r = sinh\chi$,then

$$ds^2 = a^2(\tau)(-d\tau^2 + d\chi^2 + sinh^2\chi d\Omega^2)$$

The only problem here is that we need to somehow let $sinh^2\chi \to R^2$.Thus by intuition (and because $d\Omega^2$ is more complicated),we should consider the coordinate transformation $(\tau, \chi) \to (T, R)$,such that

$$ds^2 = a^2(\tau(T, R))A^2(T, R)(-dT^2 + dR^2 + R^2 d\Omega^2)$$

By noticing $dT = \frac{\partial T}{\partial \tau}d\tau + \frac{\partial T}{\partial \chi}d\chi$ and $dR = \frac{\partial R}{\partial \tau}d\tau + \frac{\partial R}{\partial \chi}d\chi$ we see that

$$ds^2 = a^2 A^2(-(\frac{\partial T}{\partial \tau}d\tau + \frac{\partial T}{\partial \chi}d\chi)^2 + (\frac{\partial R}{\partial \tau}d\tau + \frac{\partial R}{\partial \chi}d\chi)^2 + R^2 d\Omega^2)$$

By comparing it with the first metric,we conclude that

$$A^2(-(\frac{\partial T}{\partial \tau})^2 + (\frac{\partial R}{\partial \tau})^2) = -1, A^2((\frac{\partial R}{\partial \chi})^2 - (\frac{\partial T}{\partial \chi})^2) = 1$$

$$\frac{\partial R}{\partial \tau}\frac{\partial R}{\partial \chi} = \frac{\partial T}{\partial \tau}\frac{\partial T}{\partial \chi}$$

$$A^2R^2 = sinh^2\chi$$

Plug the third equation into the first gives (Let $a, b, c, d = \frac{\partial T}{\partial \tau}, \frac{\partial T}{\partial \chi}, \frac{\partial R}{\partial \tau}, \frac{\partial R}{\partial \chi}$)

$$-a^2 + c^2 = -\frac{R^2}{sinh^2\chi}, d^2 - b^2 = \frac{R^2}{sinh^2\chi} \implies d^2 - b^2 = a^2 - c^2 = Const$$

Since by the second equation we have

$$c^2d^2 = a^2b^2 = (c^2 + Const)(d^2 - Const) \implies a^2 = c^2, b^2 = d^2 \; or \; a^2 = d^2, b^2 = c^2$$

While we can rule out $a^2 = c^2$ because $a^2 - c^2 \neq 0$.Thus

$$a = \pm d, b = \pm c \implies \frac{\partial R}{\partial \chi} = \pm\frac{\partial T}{\partial \tau}, \frac{\partial T}{\partial \chi} = \pm\frac{\partial R}{\partial \tau}$$

Since the equations are invariant under $T \to -T$,then with out loss of generality,we can let

$$\frac{\partial R}{\partial \chi} = \frac{\partial T}{\partial \tau}, \frac{\partial R}{\partial \tau} = \frac{\partial T}{\partial \chi}$$

Plug into the original equations

$$A^2((\frac{\partial R}{\partial \chi})^2 - (\frac{\partial T}{\partial \chi})^2) = 1 \implies (\frac{\partial R}{\partial \chi} + \frac{\partial R}{\partial \tau})(\frac{\partial R}{\partial \chi} - \frac{\partial R}{\partial \tau}) = \frac{R^2}{sinh^2\chi}$$

Consider the characteristic line $x = \frac{\tau+\chi}{2}, y = \frac{\chi-\tau}{2}$,then the equation becomes

$$\frac{\partial R}{\partial x}\frac{\partial R}{\partial y} = \frac{R^2}{sinh^2(x-y)}$$

I have been confused here since it is a nonlinear PDE.But I have read the reference which claims that because

$$\frac{\partial R}{\partial \chi} = \frac{\partial T}{\partial \tau}, \frac{\partial R}{\partial \tau} = \frac{\partial T}{\partial \chi}$$

In order to let $\frac{\partial^2 T}{\partial \tau \partial \chi} = \frac{\partial^2 T}{\partial \chi \partial \tau}$,we must have R satisfying wave equation

$$\frac{\partial^2 R}{\partial \tau^2} - \frac{\partial^2 R}{\partial \chi^2} = 0$$

Then R can be write in terms of

$$R = R_+(x) + R_-(y)$$

Plug in this form in to the PDE,we have

$$R'_+R'_- = \frac{(R_+ + R_-)^2}{sinh^2(x-y)}$$

For fixed $y = 0$,R_- is also fixed,thus we can integrate with respect to x

$$\int \frac{dR_+(x)}{(R_+(x) + R_-(0))^2} = -\frac{1}{R_+(x) + R_-(0)} = \int \frac{dx}{sinh^2x} = -cothx + C \implies R_+(\tau, \chi) = \alpha_1 + \frac{1}{coth\frac{\tau+\chi}{2} + \beta_1}$$

Similarly,

$$R_-(\tau,\chi) = \alpha_2 + \frac{1}{coth\frac{\chi-\tau}{2} + \beta_2}$$

Thus

$$R = \alpha + \frac{1}{coth\frac{\tau+\chi}{2} + \beta_1} + \frac{1}{coth\frac{\chi-\tau}{2} + \beta_2}$$

$$T = \alpha + \frac{1}{coth\frac{\tau+\chi}{2} + \beta_1} - \frac{1}{coth\frac{\chi-\tau}{2} + \beta_2}$$

$$A^2 = \frac{sinh^2\chi}{R^2} = \frac{sinh^2\chi}{(\alpha + \frac{1}{coth\frac{\tau+\chi}{2}+\beta_1} + \frac{1}{coth\frac{\chi-\tau}{2}+\beta_2})^2}$$

And

$$ds^2 = a^2A^2(-dT^2 + dR^2 + R^2d\Omega^2)$$

For $k = 1$,the substitution is $r = sin\chi$,and the rest has no difference,thus the coordinate transformation is

$$R = \alpha + \frac{1}{cot\frac{\tau+\chi}{2} + \beta_1} + \frac{1}{cot\frac{\chi-\tau}{2} + \beta_2}$$

$$T = \alpha + \frac{1}{cot\frac{\tau+\chi}{2} + \beta_1} - \frac{1}{cot\frac{\chi-\tau}{2} + \beta_2}$$

$$A^2 = \frac{sin^2\chi}{R^2} = \frac{sin^2\chi}{(\alpha + \frac{1}{cot\frac{\tau+\chi}{2}+\beta_1} + \frac{1}{cot\frac{\chi-\tau}{2}+\beta_2})^2}$$

And

$$ds^2 = a^2A^2(-dT^2 + dR^2 + R^2d\Omega^2)$$

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